EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 59



ECOLOGY & ENVIRONMENT-11

Why is Bio-Bitumen in the news?

- India has become the first country in the world to commercially produce bio-bitumen in road construction. The Union Transport Minister congratulated CSIR on this historic milestone. The Minister said that the initiative will help in reducing pollution from crop residue burning.
- Conventional Bitumen is a black, viscous mixture of hydrocarbons produced by the fractionation of crude oil, and it serves as a crucial binder in road construction.
- As per the Minister, bio-bitumen is a transformative step towards the vision of Viksit Bharat 2047.

What was the indigenous innovation developed by CSIR in the context of Bio-bitumen?

- It is an indigenous innovation developed by CSIR-Central Road Research Institute (CSIR-CRRI),New Delhi and CSIR- Indian Institute of Petroleum (CSIR-IIP), Dehradun.
- It involves the collection of post-harvest rice straw, its palletization and conversion into bio-oil through pyrolysis, which is then blended with conventional bitumen.
- Extensive laboratory tests have shown that 20-30% of conventional bitumen can be safely replaced without compromising the performance.

List out various important aspects about Bio-bitumen.

- Bio-bitumen is manufactured using renewable organic materials, such as plant-based oils, agricultural waste, or biomass. It can be obtained from vegetable oils, crop stubble, algae, lignin, animal manure. These materials undergo a special processing method to create a high-quality binder that is similar to traditional bitumen.
- Raw materials, such as plant-based oils, lignin or algae are collected and pre-processed. Thermal decomposition of biomass at controlled temperatures produces bio-oil, which serves as a precursor for bio-bitumen. Bio-oil undergoes refining and polymer modification to enhance its viscosity, thermal stability, and adhesive properties.
- In some cases, bio-bitumen is blended with conventional bitumen to improve performance characteristics, while maintaining sustainability benefits.

What are the important aspects that emanated in the context of wild animals destroying crops?

In 2025, the National Board of Wildlife recommended reinstating protection for the rhesus macaque as a Schedule-II animal.

- Wild animals are destroying crops and adding massively to the economic losses of farmers. In many parts of India, farming is becoming unviable because of marauding animals. There is now a war involving not only big animals such as tigers, elephants and leopards, but also wild boars, nilgai and other ungulates and most damaging of all, monkeys.
- Recently, the Centre included animal related losses in the National Crop Insurance Scheme, the Pradhan Mantri Fasal Bima Yojana.
- Farmers are given a 72-hour window to report the loss which will be verified using drones before compensation is paid. But as per a prominent survey, even when 25% of farmers affected by wild animal attacks sought compensation, only 1-2% received payments commensurate with the damage.
- Under Section 62 of the Wildlife Protection Act, only the Centre can declare a wild animal vermin, which would then allow it to be culled by the Forest Department.
- Recently, Kerala government introduced its own Wildlife Protection Bill declaring wild boar attacks a statewide disaster and classifying the animal as vermin. However, the bill is with the President.

What are the natural factors that lead to land subsidence?

- The land subsidence may occur due to tectonic activity that include fault movement, regional geological settling, and natural compaction of deltaic and coastal sediments over time.
- Glacial isostatic adjustment, where land continues to settle following the retreat of ice sheets thousands of years ago can be another reason.
- Geological conditions include natural consolidation of soft, recently deposited sediments.

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What are the anthropogenic activities that accelerate land subsidence?

- While natural processes cause subsidence, anthropogenic activities often accelerate background subsidence. For example, excessive groundwater extraction causes aquifer systems to compact, particularly clay layers that act like sponges when water is removed.
- Other processes include oil and gas extraction and mining.
- While natural subsidence typically occurs at rates of millimeters per year, human-induced subsidence can reach several centimeters annually.

List out various environmental issues in the news.

- Pampadum Shola National Park is in the news for clearing Australian wattle, an invasive species, in certain areas in Kerala. The forest department had introduced the Australian wattle praised for its tannin-rich bark used in leather processing at Pampadum in the 1960s, along with eucalyptus and pine.
- West Bengal confirmed presence of Nipah virus after 19 years. It is bat-born and can cause fever, seizures and encephalitis in humans.
- Kumbhalgarh Wildlife Sanctuary was declared an eco-sensitive zone by the Union Government and banned activities such as mining and stone quarrying in 243 square km. It is in the Aravalli Hills region of Rajasthan.
- Mizoram officials said 20 wild boars were found to have died after contracting African Swine Fever (ASF). The deaths caused concern about potential spread of the ASF virus among pigs.

What are the environmental concerns expressed in the context of Great Nicobar Island project?

- It is one of India's most biologically distinctive and culturally sensitive regions.
- It hosts several vulnerable species, including the leatherback sea turtle, robber crabs, Nicobar megapode (a bird endemic to the islands), and the Nicobar long-tailed macaque, another endemic species. The largest living reptile, the saltwater crocodile roams these islands as the apex predator.
- These are not just peripheral species, they are ecological sentinels of beaches, mangroves, and the primary rainforest systems that have evolved over millennia in relative isolation.
- The archipelagic island chain is also home to The Shompen, a Particularly Vulnerable Tribal Group (PVTG) and the Nicobarese community, whose lives are interwoven with forests and coasts.

Small steps

India's third set of nationally-determined contributions (NDCs) makes small upgrades to the last one

	NDC-1	NDC-2	NDC-3	PROGRESS
Year of announcement	2015	2022	2026	
Target year	2030	2030	2035	
Goal 1: Reducing emissions intensity (emissions per unit GDP)	**33-35%** from 2005 levels	**45%** from 2005 levels	**47%** from 2005 levels	**36%** reduction achieved by 2020
Goal 2: Share of non-fossil electricity sources	40%	50%	60%	**52%** in February 2026
Goal 3: Creating additional carbon sinks (forests, trees)	**2.5 to 3 bn tonnes** over 2005 stock	**2.5 to 3 bn tonnes** over 2005 stock	**3.5 to 4 bn tonnes** over 2005 stock	**2.3 bn tonnes** created by 2021

India and Argentina were the only G20 nations that had not announced a 2035 NDC by the end of 2025. Incidentally, India's CO_2 emissions in 2025 grew at the slowest rate in more than two decades.

UNION CABINET APPROVED INDIA'S UPDATED NDCs

- India committed 60% of its installed electric capacity from non-fossil sources by 2035, 47% reduction in intensity of emissions per unit of GDP (from 2005 levels) and increasing its carbon sink to 3.5 billion to 4 billion tons of CO_2 equivalent. These targets will be communicated to the UNFCCC.
- Under the Paris Agreement, every signatory country must periodically submit NDCs, which are voluntary pledges spelling out how they will transition away from fossil fuels.
- India's previous NDC, conveyed in August 2022, committed it to 50% non-fossil installed capacity by 2030, a 45% reduction in emissions intensity, and a carbon sink of at least 2.5 billion-3 billion tonnes of CO_2-equivalent. As of early 2026, about 52% of India's installed capacity comes from non-fossil sources.

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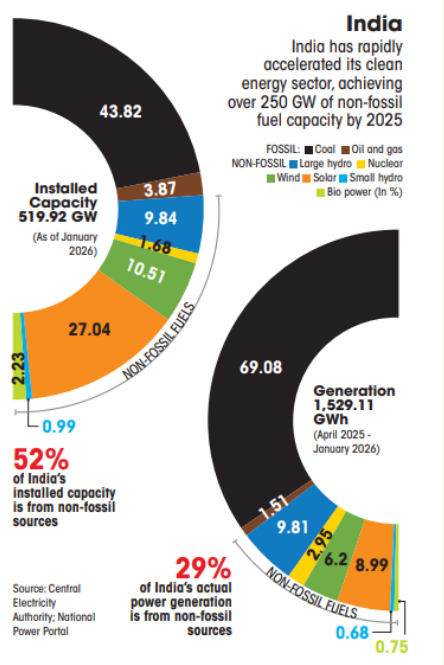
HENCE, WE ARE NOT WORKING TOWARDS UAE CONSENSUS!

Learning Space

- UAE Consensus was an agreement at COP28 by nearly 200 countries in December 2023 to accelerate climate action by transitioning away from fossil fuels, tripling global renewable energy capacity and doubling energy efficiency improvements by 2030 to limit warming to 1.5 degree Celsius.
- However, no country set a target for winding down oil and gas production. Nearly three quarters did not mention fossil fuel subsidy reform and most developing country NDCs depend on international finance, which is awfully short of the required scale.

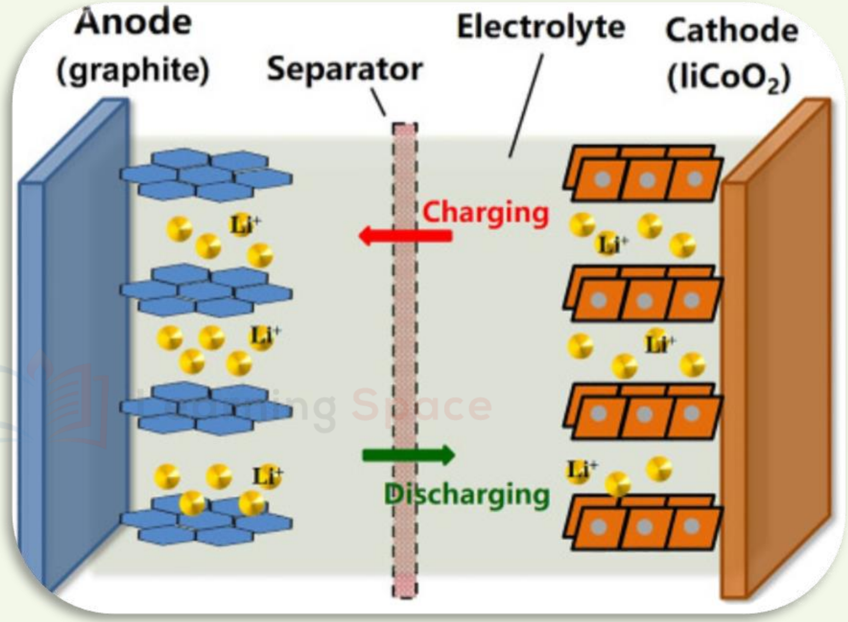
However, green shoots are emerging through solar and wind installations that reached a record 814 gigawatt in 2025. Renewables surpassed coal as the largest source of electricity globally in the first half of 2025.

FROM INDIA, THERE IS DRAMATIC DECELERATION IN THE CO_2 EMISSIONS

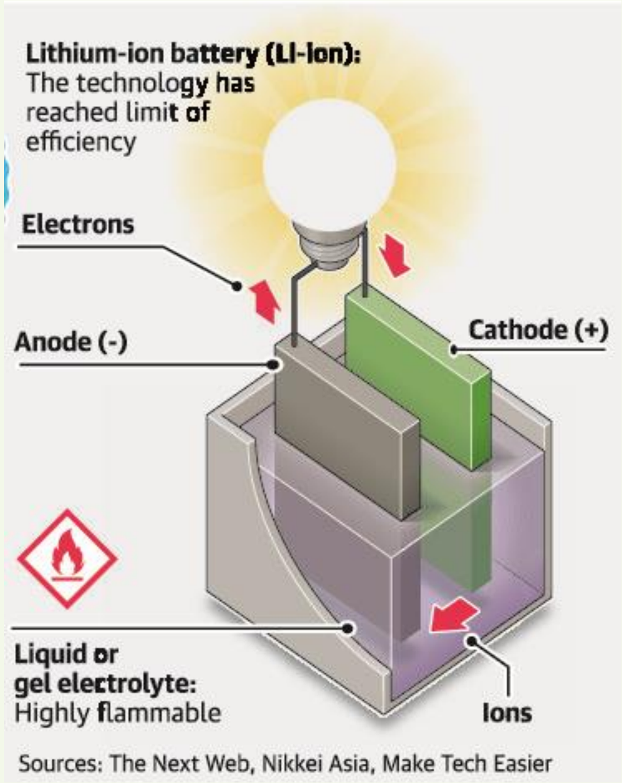


- As per Centre for Research on Energy and Clean Air (CREA), India's CO_2 emissions grew by just 0.7% in 2025, the slowest rate since 2001, excluding the Covid year of 2020. This is a dramatic deceleration from 4 to 11% growth in 2021-24.
- The power sector was the key driver - emissions fell 3.8%, after coal-fired generation declined for the first time outside of Covid since 1973.
- India added 47 GW of solar, 6.3 GW of wind, 4 GW of hydro, and 0.6 GW of nuclear in 2025, or enough new clean generation to cover demand growth of up to 5%.

WHAT ARE VARIOUS COMPONENTS THAT GOES INTO THE MANUFACTURING OF LITHIUM-ION BATTERY?



- Every Lithium-ion battery consists of three active components:
  - ✓ The anode, typically graphite
  - ✓ The cathode, typically based on a nickel, cobalt, and manganese-based oxide and
  - ✓ An electrolyte, typically salts of lithium in an inorganic solvent.
- The Battery manufacturing is a complex operation involving forming the sheets of the anode and cathode and assembling them into a sandwich structure held apart by a thin separator.
- Separators, about 15 microns in thickness - about a fifth of the thickness of the human hair - perform the critical function of preventing the anode and cathode from shorting.



QUALITY IS THE MOST IMPORTANT ASPECT IN THE BATTERY MANUFACTURING

- Various layers are assembled with high precision with tight tolerances.
- Safety features, such as thermal switches that turn off if the battery overheats, are added as the sandwich is packaged into a battery cell.
- Battery cells are assembled into modules and then further assembled into packs.

LET US UNDERSTAND THE DYNAMICS OF THESE THREE ELEMENTS!

1 HEAT

- If an adverse event such as a short circuit occurs in the battery, the internal temperature can raise as the anode and cathode release their energy through the short.
- This can lead to a series of reactions from the battery materials, especially the cathode, that release heat in an uncontrolled manner, along with oxygen.
- They rupture the seal of the battery.

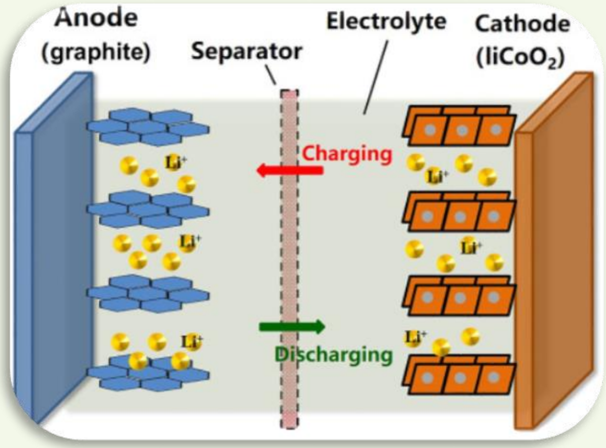
2 OXYGEN

When the seal of the battery is ruptured, it further exposes the components to outside air and the second part of the fire triangle, namely, oxygen comes into picture.

3 FUEL

The liquid electrolyte, which is flammable, serves as a fuel.

The combination leads to a catastrophic failure of the battery resulting in smoke, heat, and fire, released instantaneously and explosively!



WHAT ARE THE REASONS FOR THERMAL RUNAWAY OF EV BATTERIES (EV BATTERY FIRES)?

Petrol cars too catch fire. The difference is that EV battery fires burn hotter, spread faster and are harder to put out, as the battery releases oxygen as it burns.

1. A lithium-ion battery packs thousands of cells tightly together, each generating heat as it charges and discharges. Normally, an onboard computer called the Battery Management System (BMS) keeps the temperature within a safe range. If something goes wrong, one cell can overheat, triggering a chain reaction.
2. Moreover, toxic gases are released in the process, including hydrogen fluoride, a flammable vapour.
3. Battery packs are packed inside shells of reinforced steel or aluminium, but strong impact, for example, to the undercarriage, can deform the casing and puncture the cells inside, leading to a short circuit.
4. Charging a battery beyond its designed capacity can force charge to build up in the wrong places inside the cells.

WHAT ARE THE REASONS FOR THERMAL RUNAWAY OF EV BATTERIES (EV BATTERY FIRES)?

Learning Space

5. As the battery expands and contracts during use, rare manufacturing defects, like a small protrusion of metal, can bring the positive and negative electrodes in touch, causing an enormous current to flow between them.
6. In hot weather, like during summers, the cooling system can struggle to shed heat. Parking an EV in direct sunlight for long periods or charging it immediately after a long drive can add to the thermal stress.
7. Extension cords or domestic wiring in old buildings can also overheat when they can't handle the sustained current.
8. As batteries age, their internal components also degrade. So, users who ignore warning lights or skip inspections can miss early signs of swelling or chemical decomposition.
9. The contaminated water, after flooding or after heavy rains, can infiltrate a battery pack and cause short circuits.

WHAT ARE THE STEPS TAKEN BY THE EV INDUSTRY?

- Most EVs today have channels alongside the cells filled with a coolant that absorbs their heat and dissipates it into the air.
- Scientists are currently developing a new form of cooling, where the coolant evaporates as it absorbs heat and releases it into the air, improving heat transfer and handling temperature spikes better.
- Manufacturers are also exploring batteries that use a solid electrolyte rather than the current liquid, reducing the risk of a thermal runway.
- Firewalls are being refined inside existing designs so that if one cell fails, the fire doesn't spread.
- The BIS released updated safety norms for EV batteries in 2023 after a spate of fires. As part of this, the Automotive Research Association of India also requires tests to check how heat propagates in a battery.
- Rules also require battery packs to give a vehicle's users at least five minutes to escape before a fire.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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